

DRY-RUN GUIDE | 11_classify_age.cpp
Source: cpp\1st_belt\11_classify_age.cpp

CORE BEHAVIOR

classifyAge returns one of four age labels. The ordered comparisons cover children 0-12, teens through 19, adults through 64, and seniors above 64. Negative ages fall through to Teenager in this implementation.

REPRESENTATIVE INPUT / CALL

Input: '20'.

STEP TRACE

age>=0 && age<=12 is false; age<=19 is false; age<=64 is true, so return Adult.

EXPECTED OUTPUT / RETURN

'Adult'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program that takes the age of a person as input and classifies
004 | them into one of the following age groups:
005 |
006 | 0-12: "Child"
007 | 13-19: "Teenager"
008 | 20-64: "Adult"
009 | 65 and above: "Senior"
010 |
011 | Input Format:
012 | A single integer representing the age of the person.
013 |
014 | Output Format:
015 | A single line containing the age group classification as a string:
016 | "Child", "Teenager", "Adult", or "Senior".
017 | */
018 |
019 | #include <iostream>
020 | #include <string>
021 | using namespace std;
022 |
023 | string classifyAge(int age) {
024 |     if (age >= 0 && age <= 12) {
025 |         return "Child";
026 |     } else if (age <= 19) {
027 |         return "Teenager";
028 |     } else if (age <= 64) {
029 |         return "Adult";
030 |     } else {
031 |         return "Senior";
032 |     }
033 | }
034 |
035 | int main() {
036 |     int age;
```

```
037 |     cin >> age;
038 |
039 |     cout << classifyAge(age);
040 |
041 |     return 0;
042 | }
043 |
```